

5/6/2024

Informatics 3 Year Project

Deliverable 3: Architecture Design

CYBER CREW



TEAM NAME: CYBER CREW

TEAM NUMBER: 4

PROJECT NAME: AUTOWORX BARNYARD ONLINE

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1. Updated Problem Statement, Proposed Solution & Use Cases

1.1 Problem Statement

The current AutoWorx Barnyard's customer communication and inquiries, invoice & quotation generation, task coordination, and process documentation rely on outdated methods which include the use of telephone & WhatsApp for communication, Microsoft Word Invoice & Quotation Template, task assignment by mouth resulting in a lack of systematic customer relationship management, task delegation, and process documentation hindering productivity, customer engagement, and knowledge sharing.

1.2 Proposed Solution

We will develop a workshop management system that prioritizes customer engagement, seamless communication, and efficient service estimation costs. This system aims to improve operations by enhancing transparency, delegating mechanics' tasks, automating the quoting and invoice generation process, maintaining comprehensive client service history, and promoting skill sharing among team members to ensure consistent service quality.

1.3 Use Cases

1.3.1 User Management

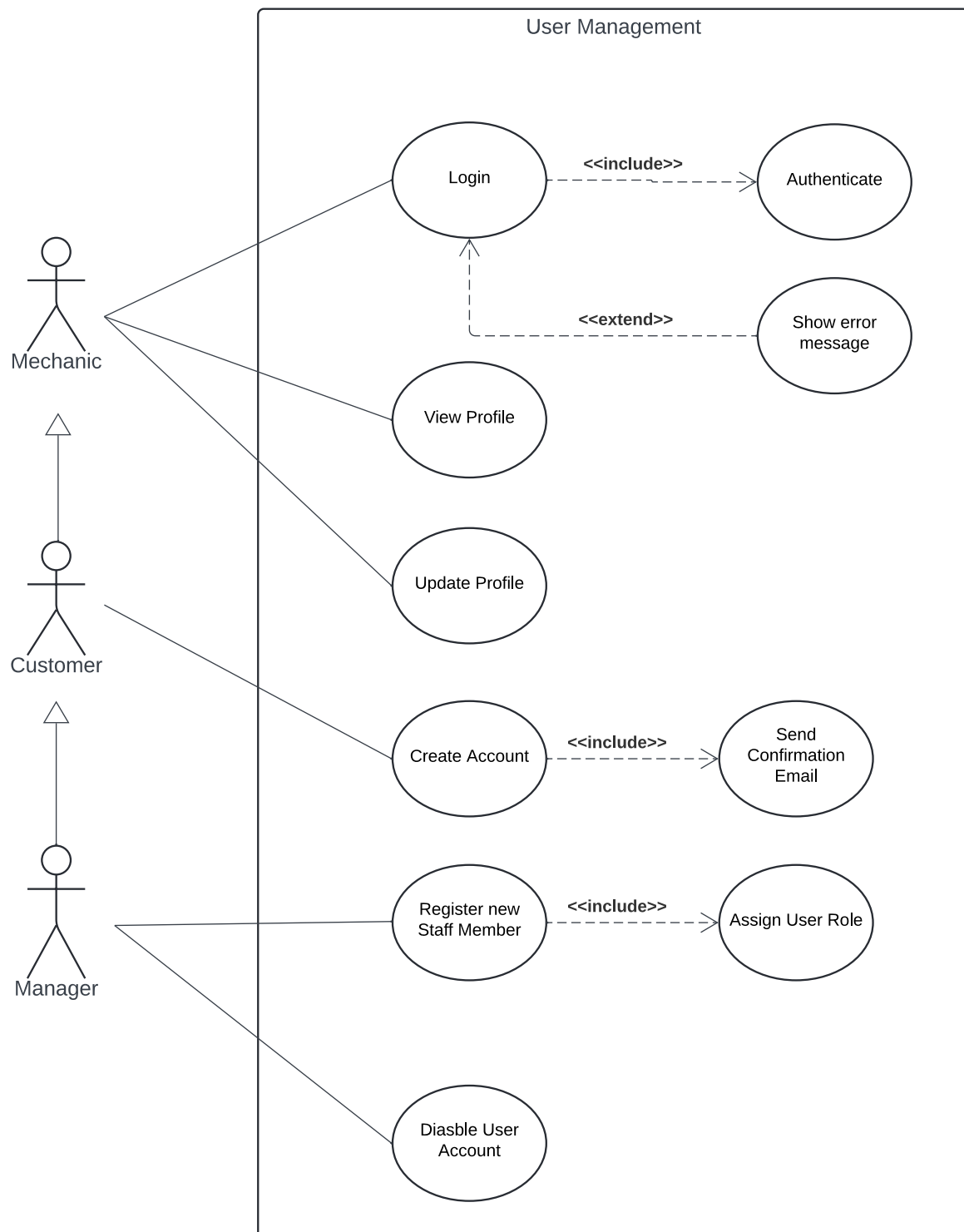


Figure 1: User Management Use Case Diagram

1.3.2 Service Request Management

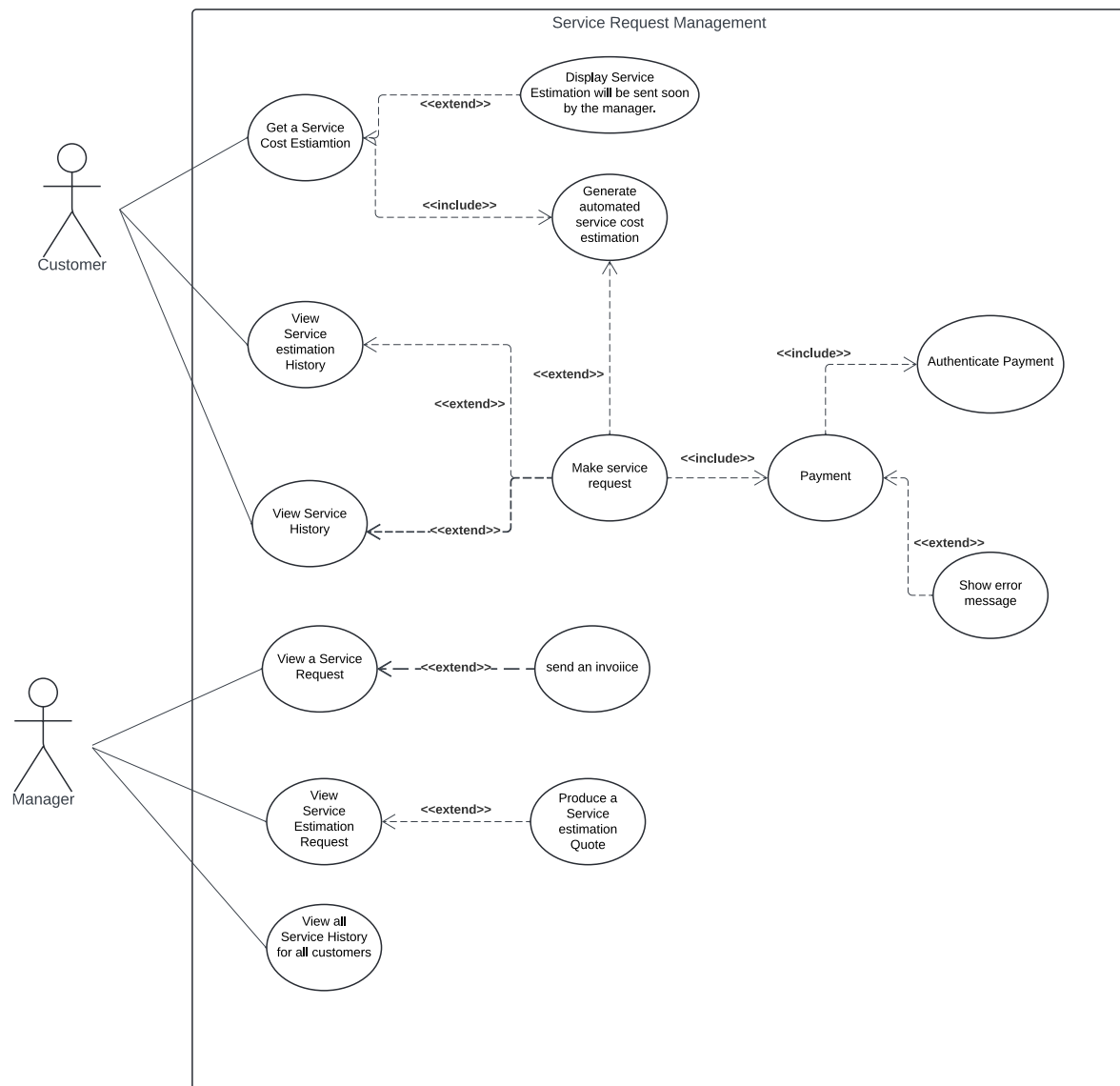


Figure 2: Service Request Management Use Case Diagram

1.3.3 Task Management

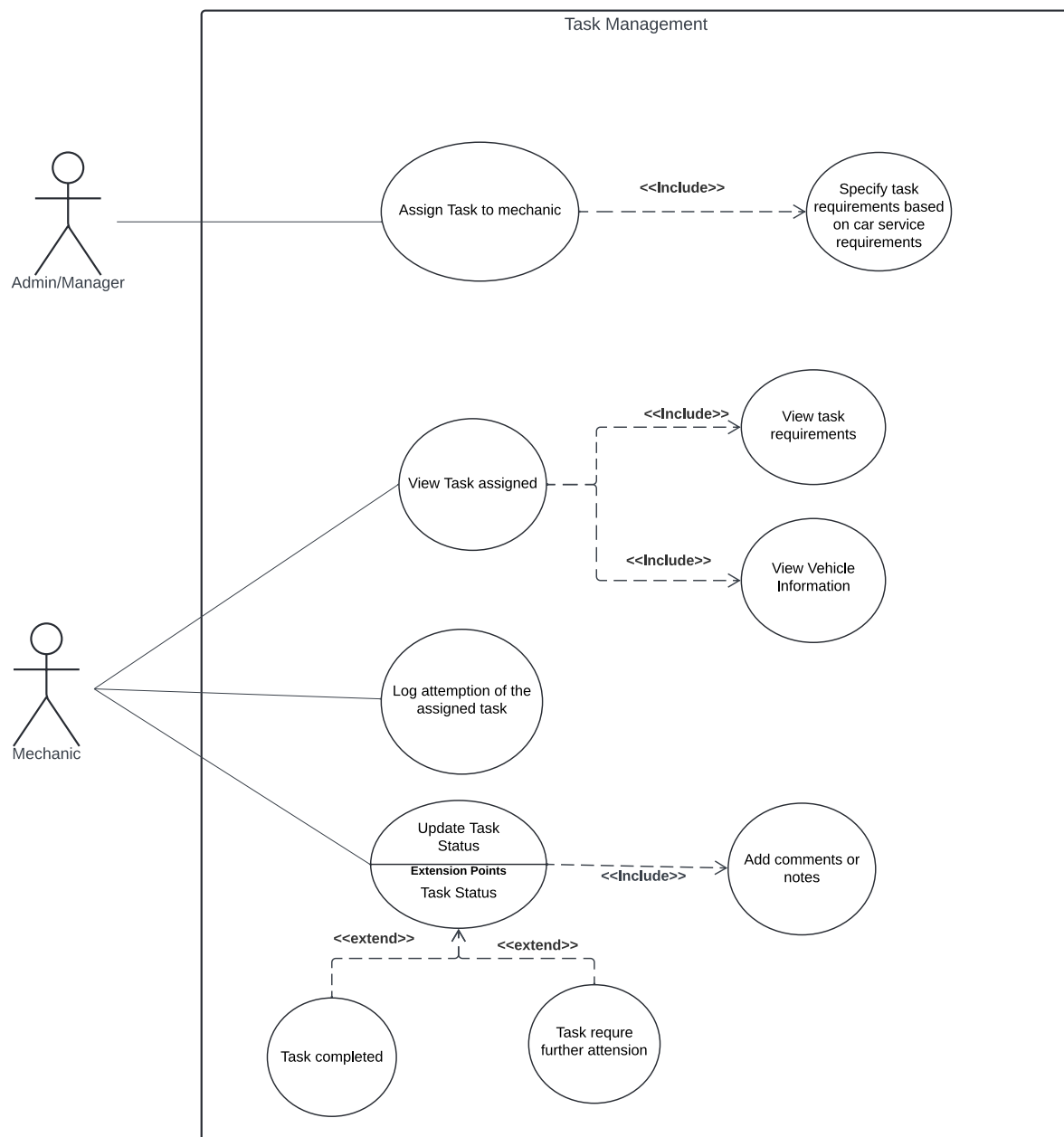


Figure 3: Figure 3: Task Management Use Case Diagram

1.3.4 Performance Monitoring

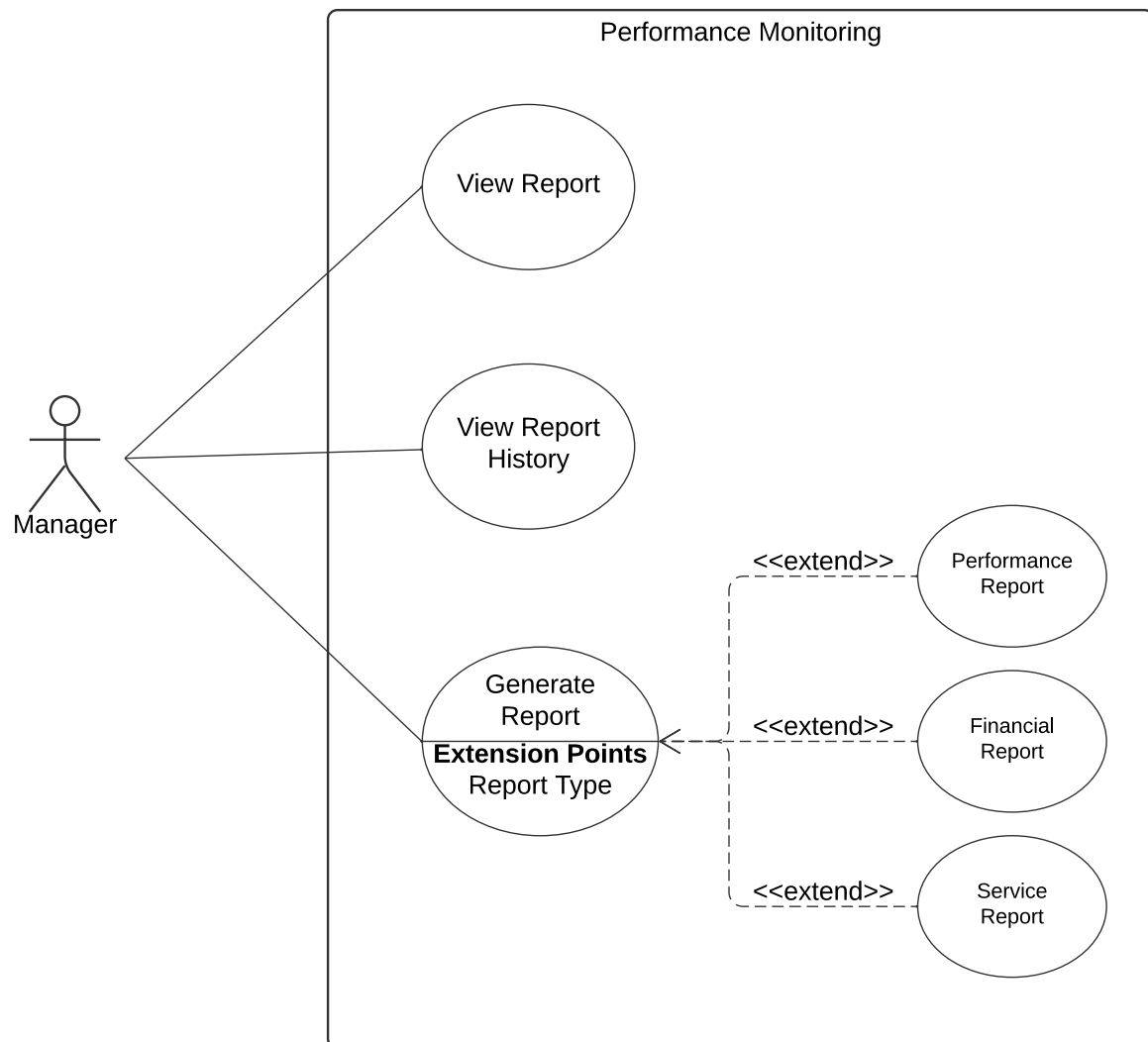


Figure 4: Skills Sharing Use Case Diagram

1.3.5 Appointment Management

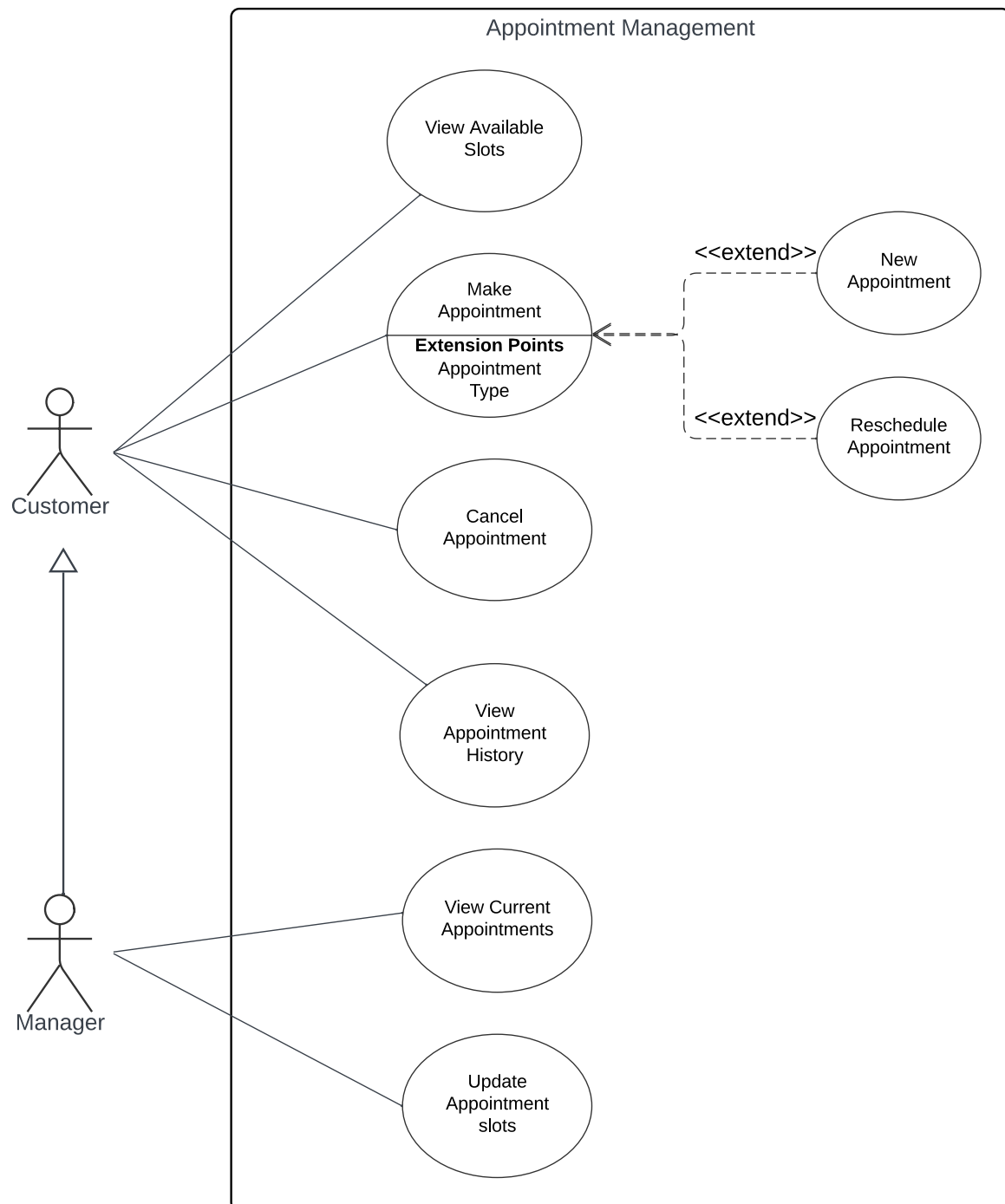


Figure 5: Appointment Management Use Case Diagram

1.3.6 Skills Sharing

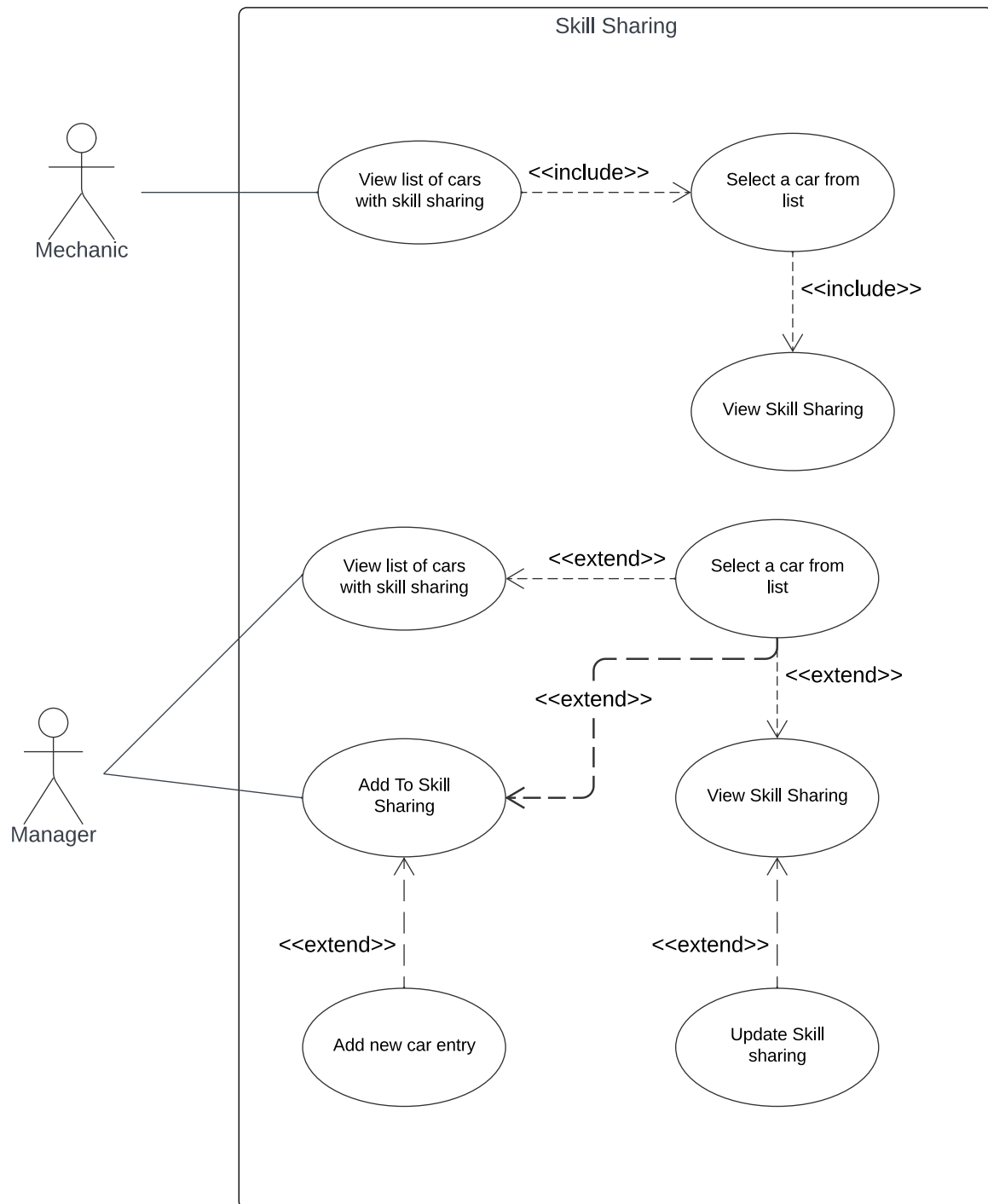
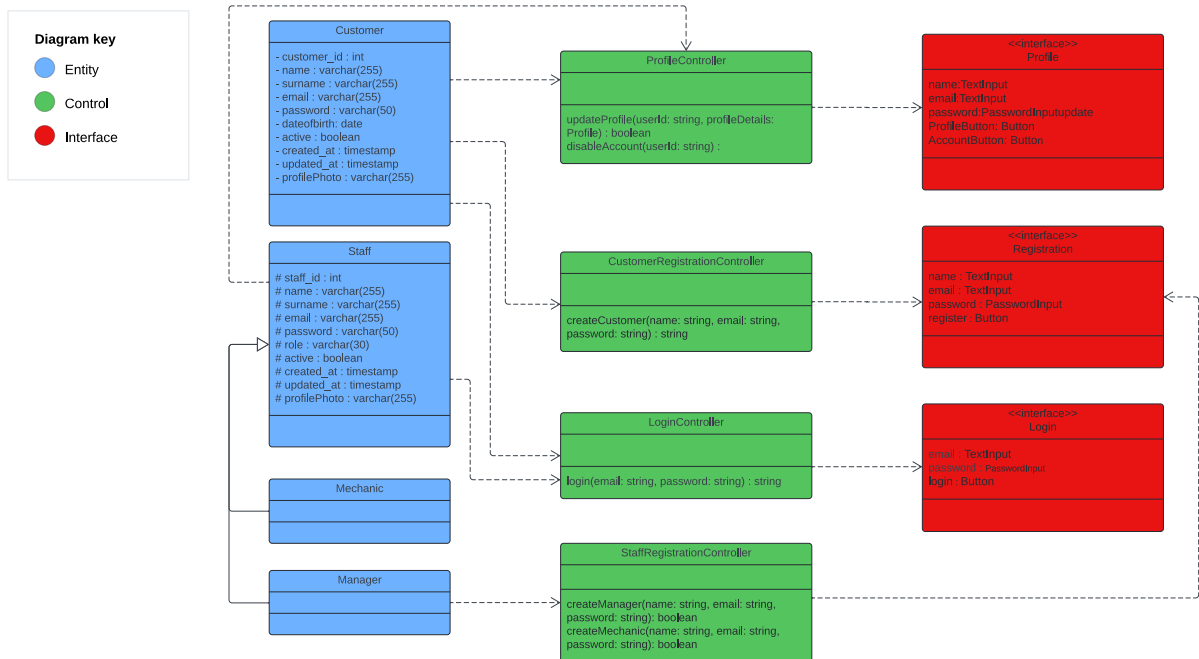


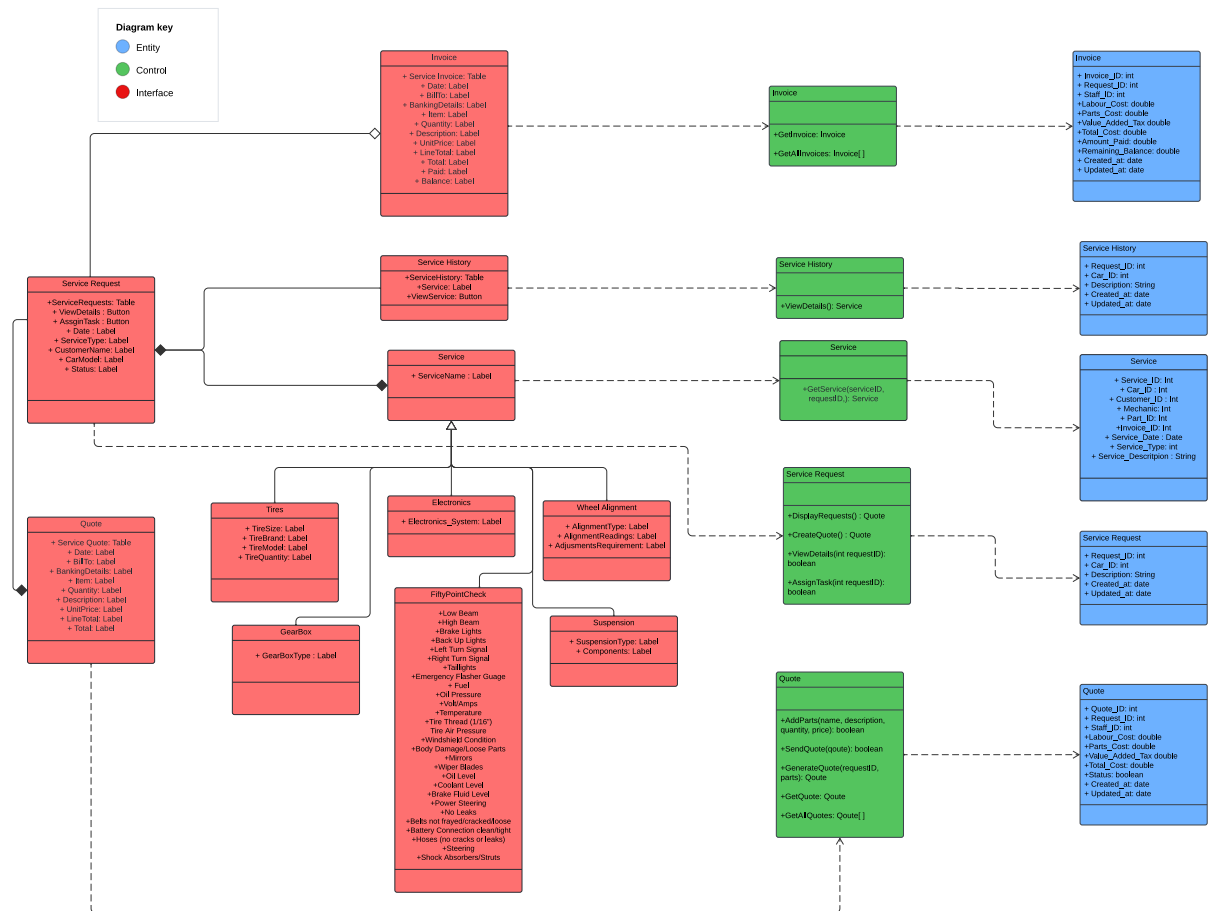
Figure 6: Skills Sharing Use Case Diagram

2. Class Diagrams

2.1 User Management



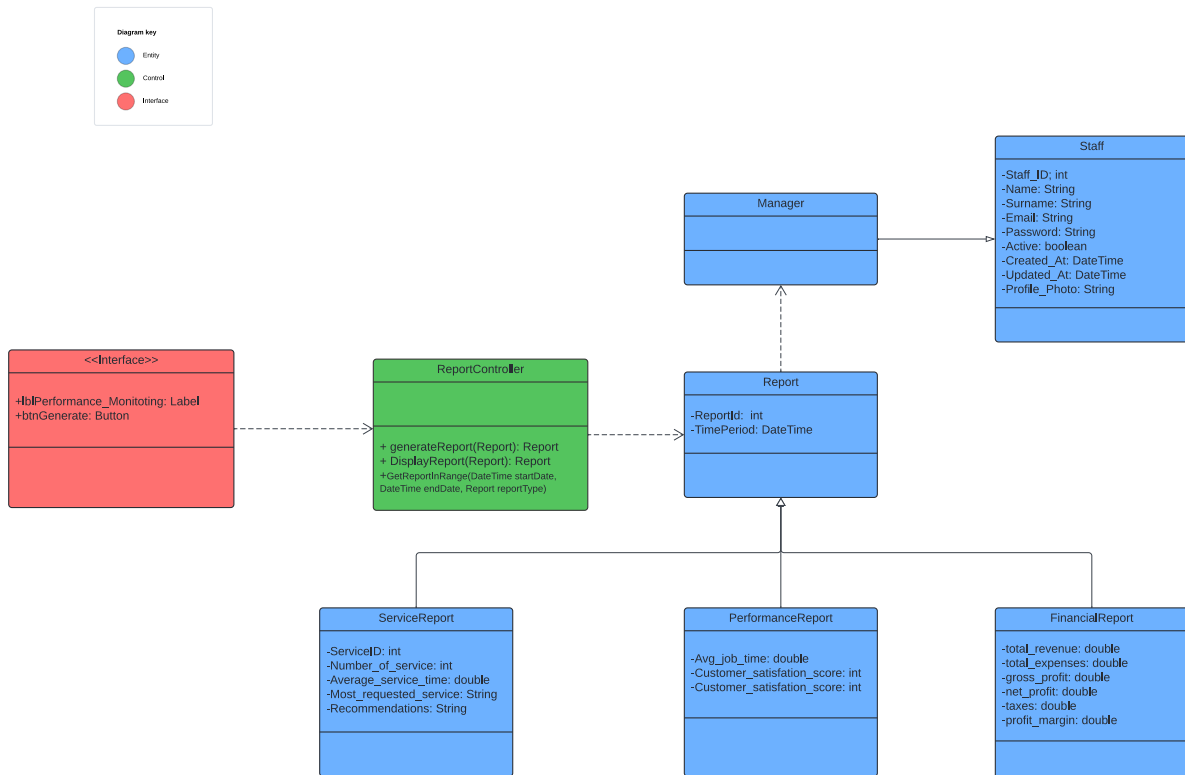
2.2 Service Request Management



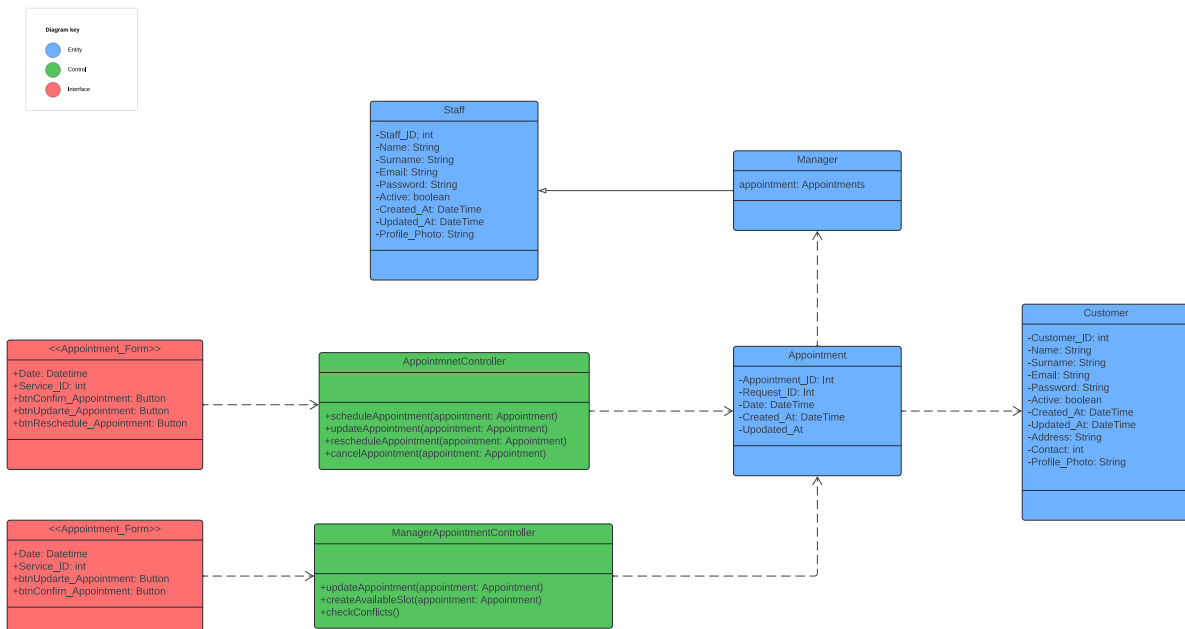
2.3 Task Management



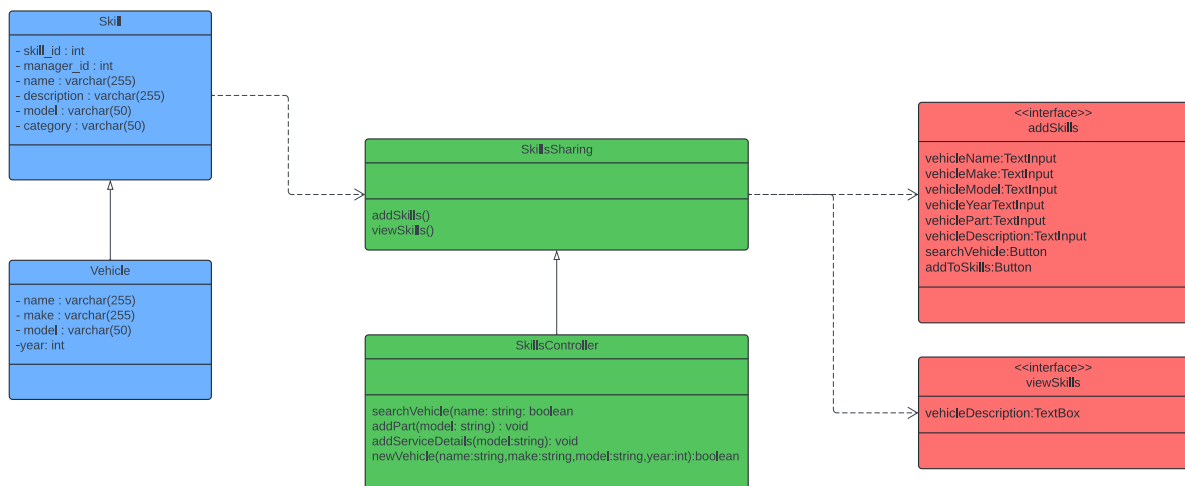
2.4 Performance Monitoring Management



2.5 Appointment Management



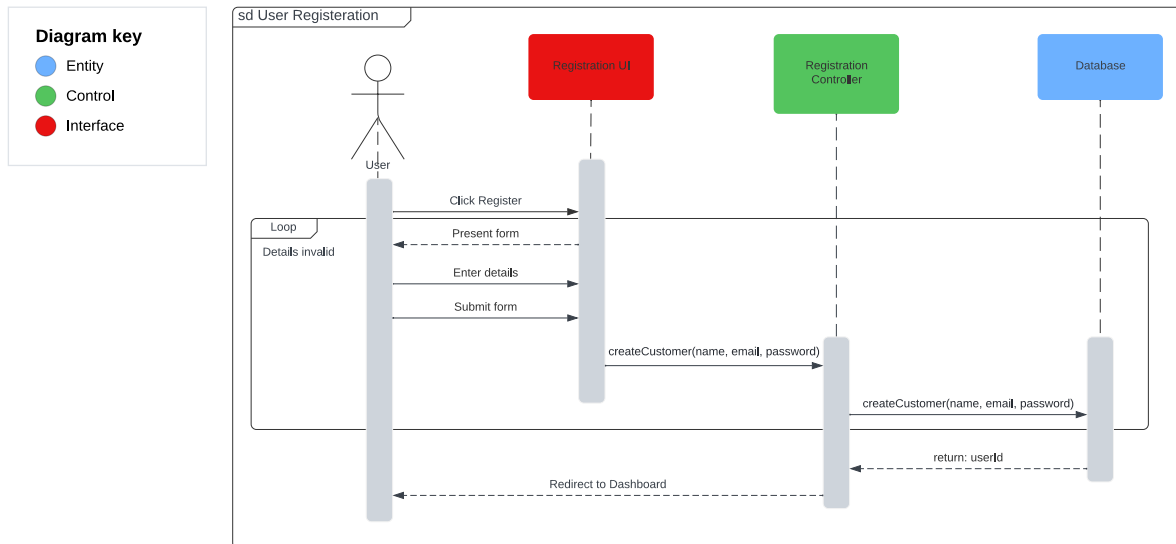
2.6 Skills Sharing



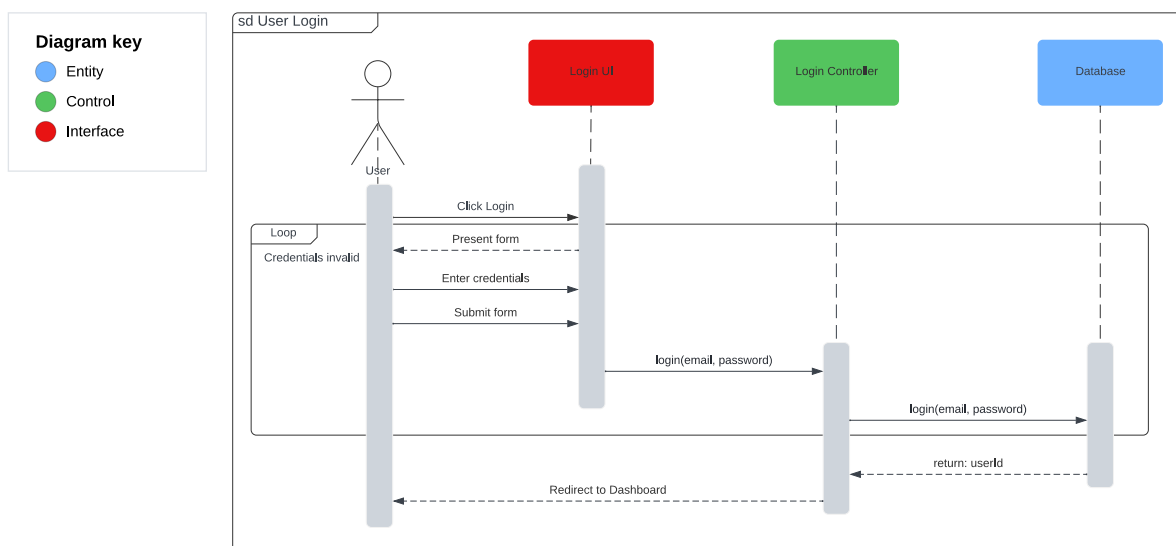
3. Interaction Sequence Diagram

3.1 User Management

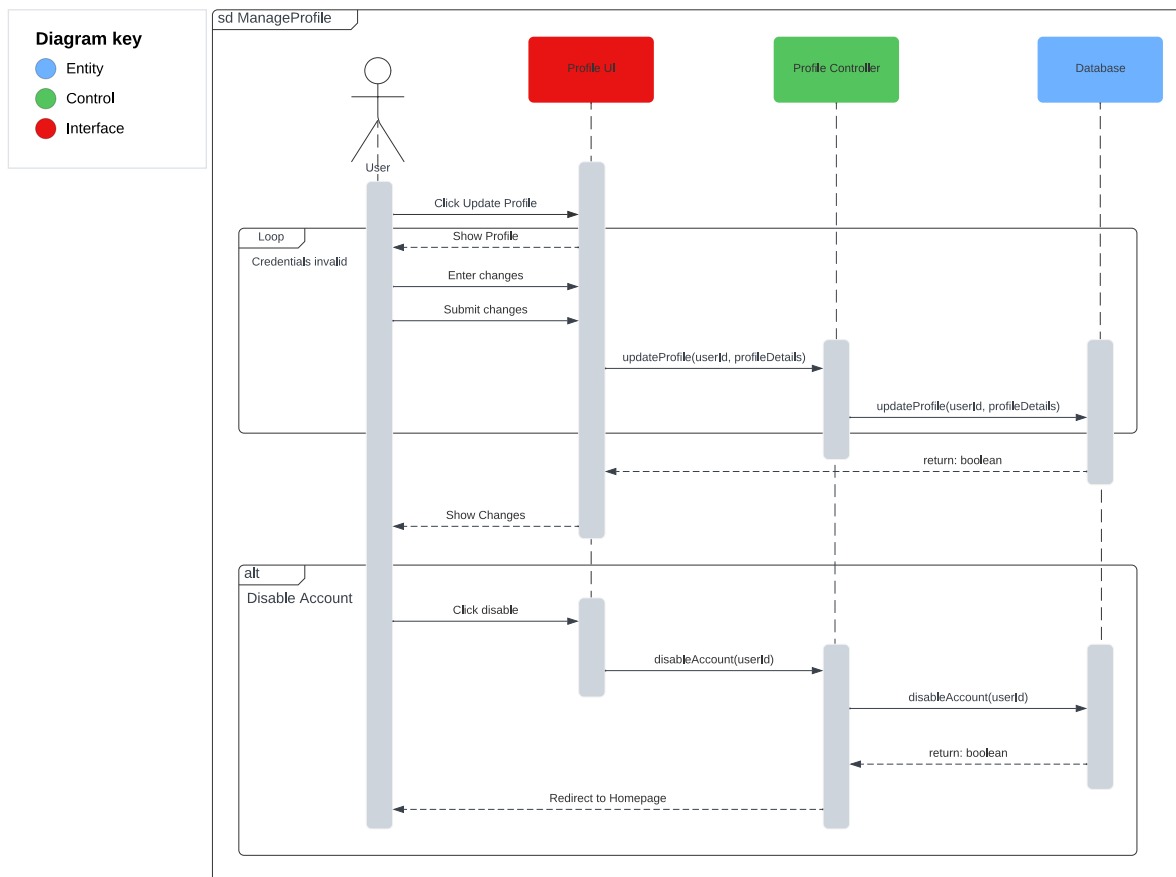
a) Registration



b) Login



c) Manage Profile

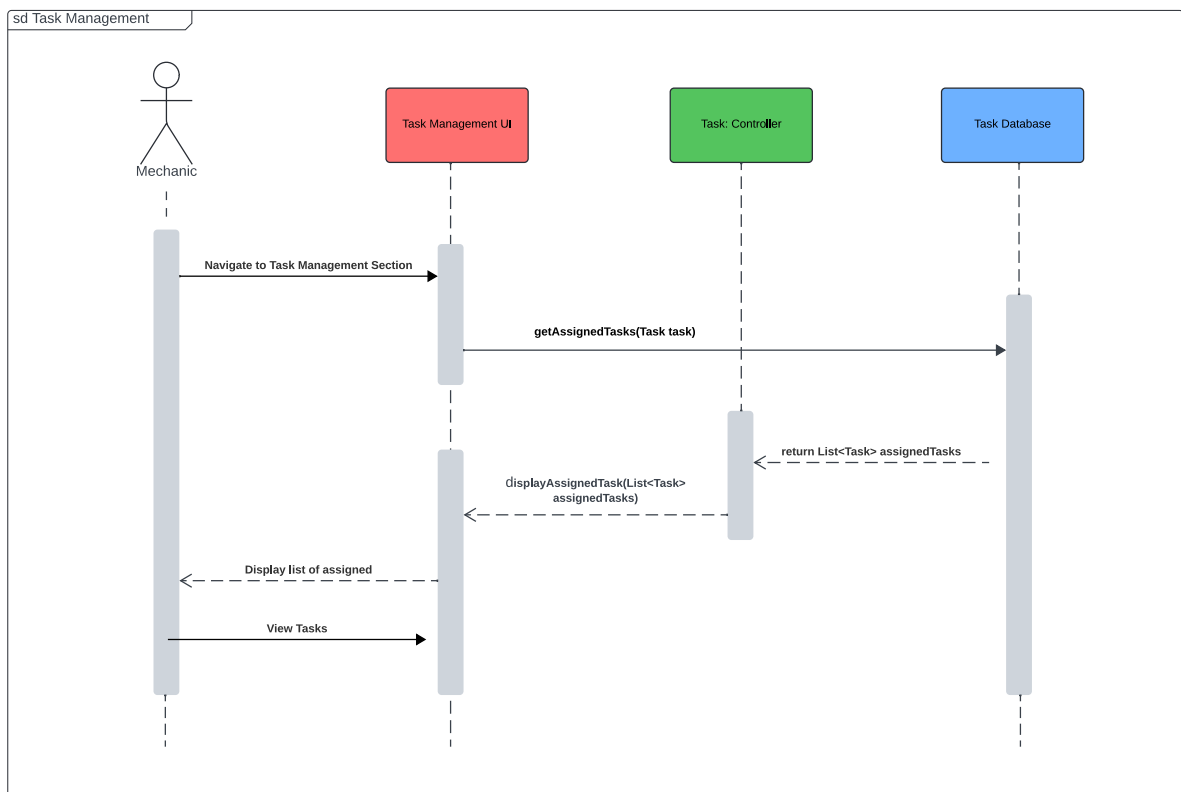


3.2 Service Request Management

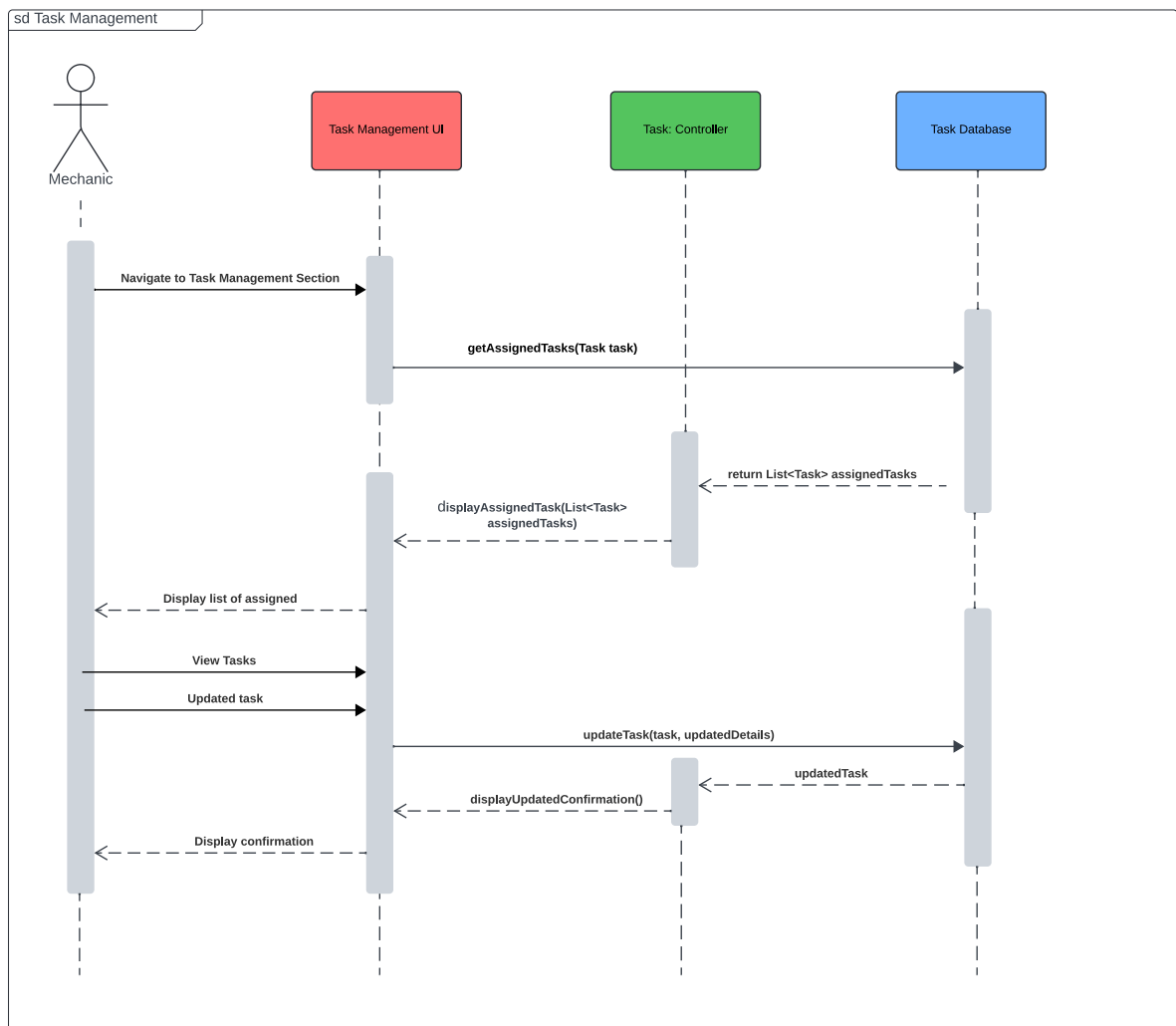


3.3 Task Management

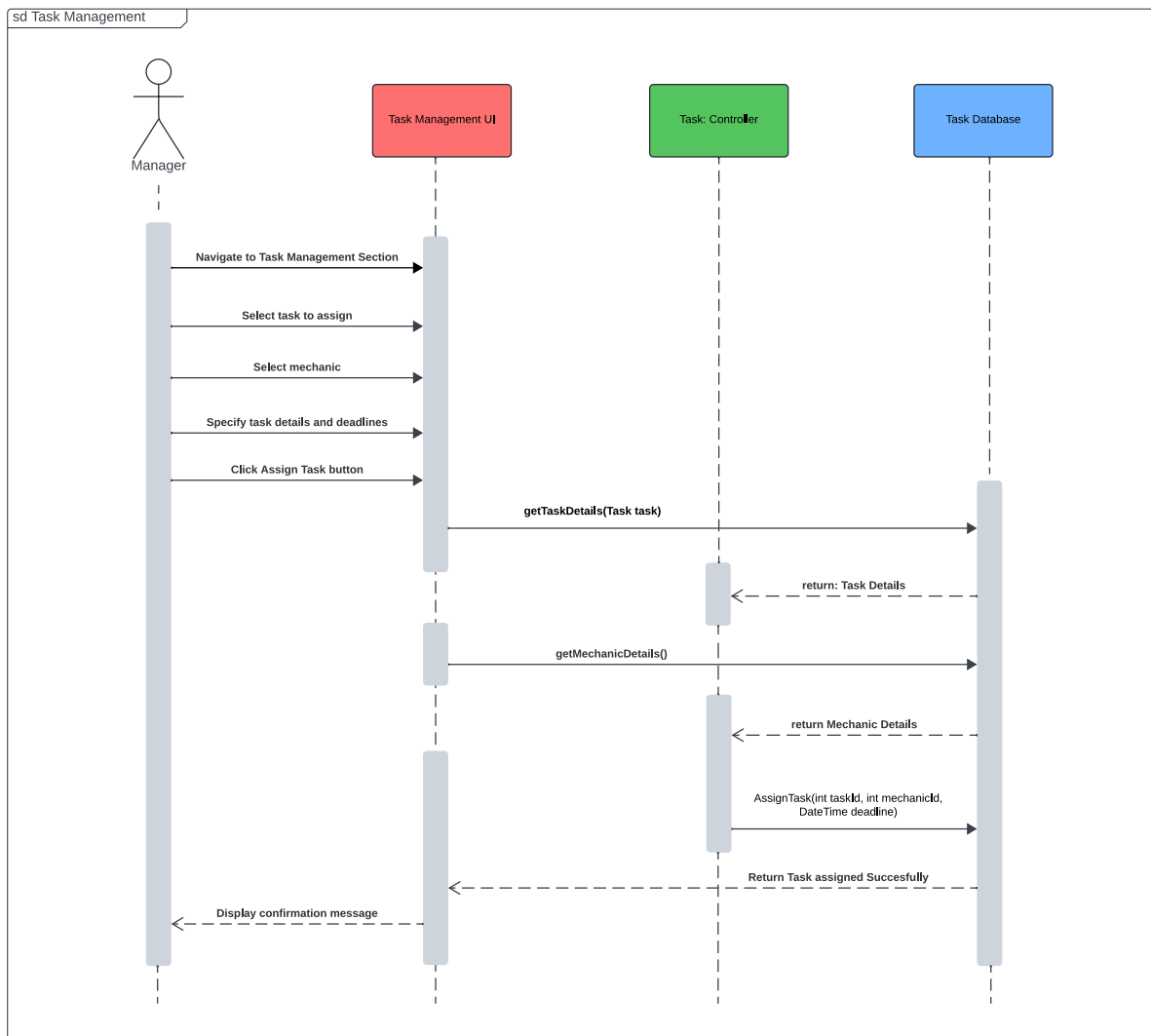
a) View Task



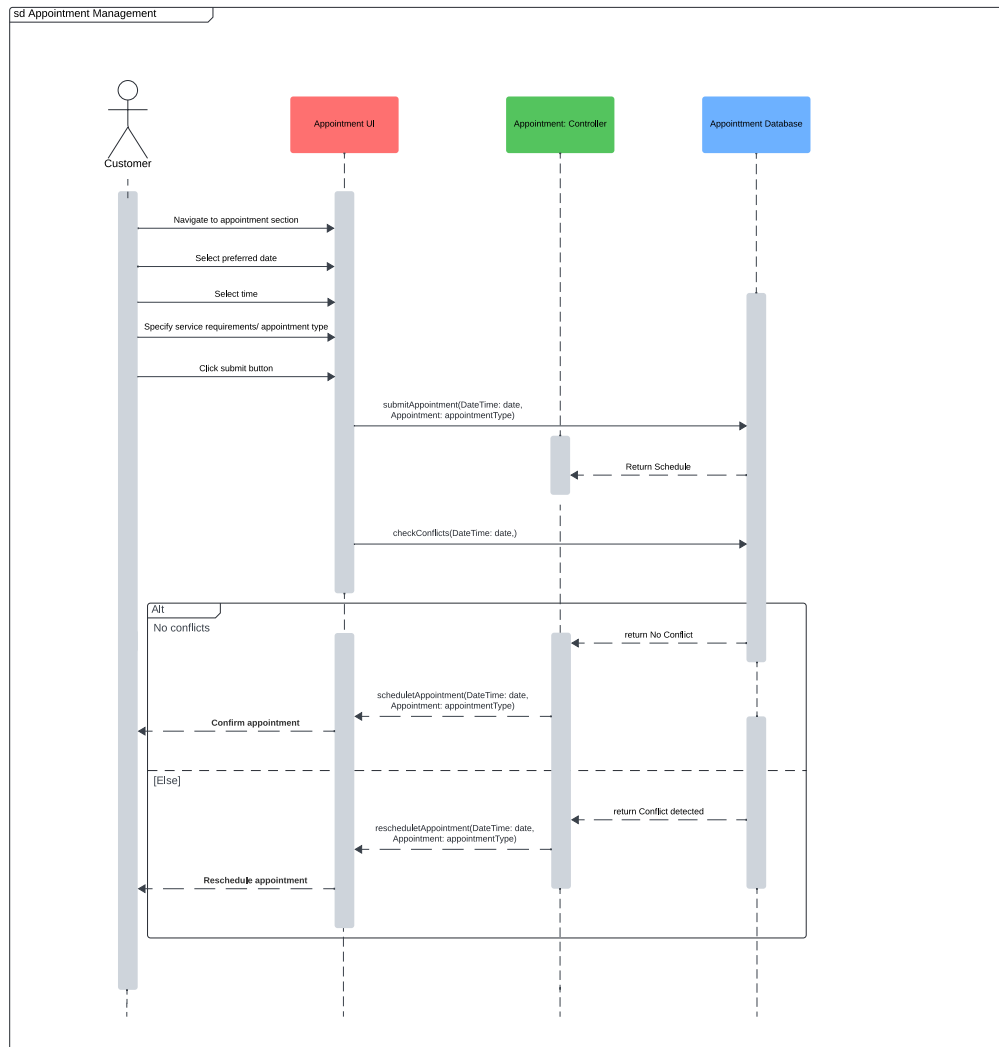
b) Update Task



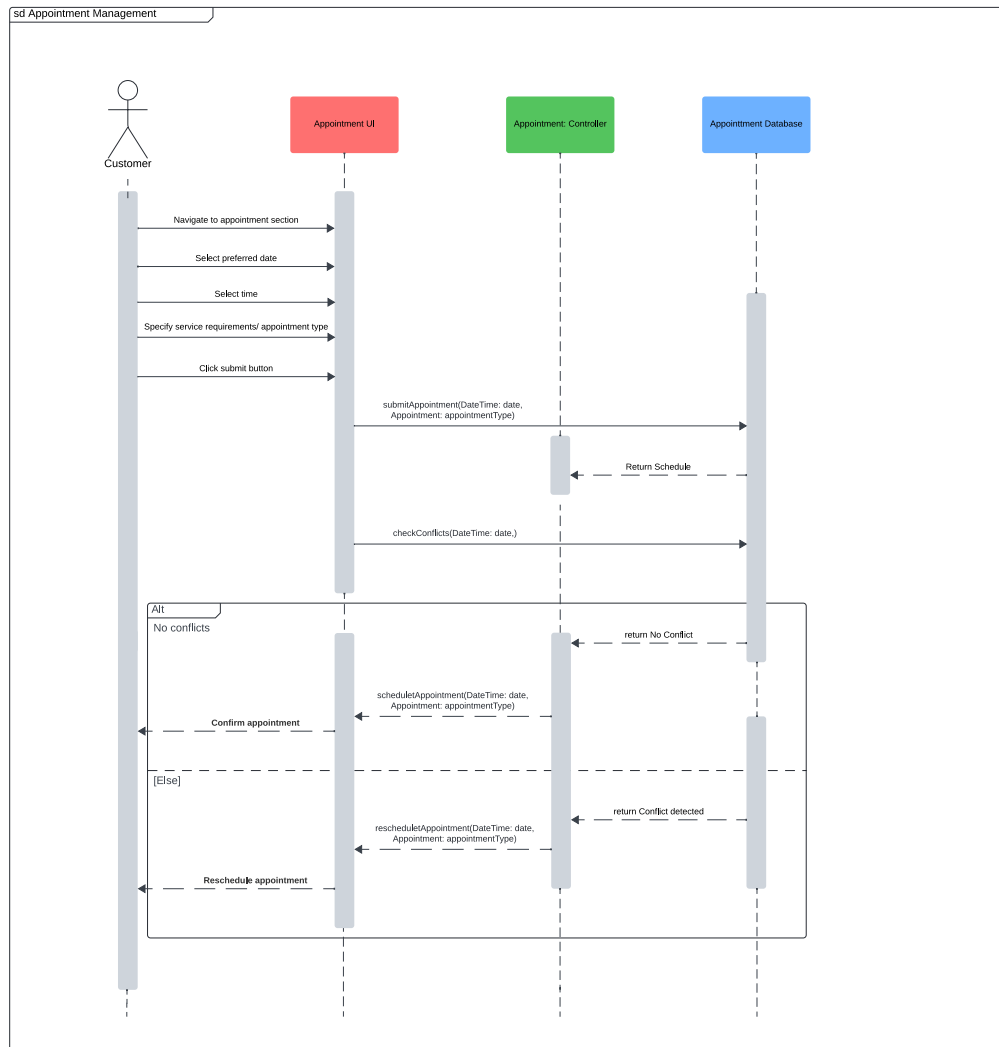
c) Assign Task



3.4 Performance Monitoring Management

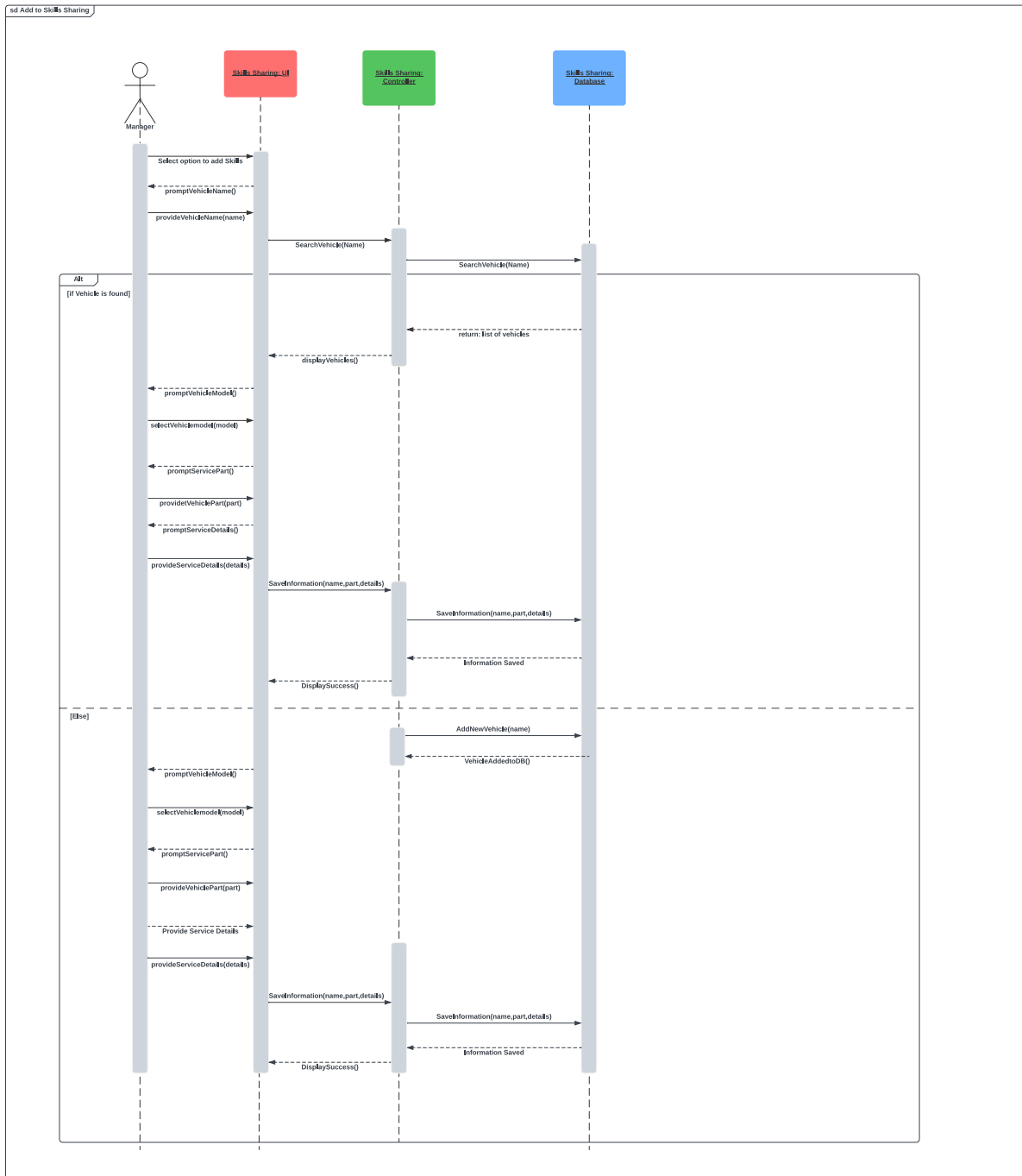


3.5 Appointment Management

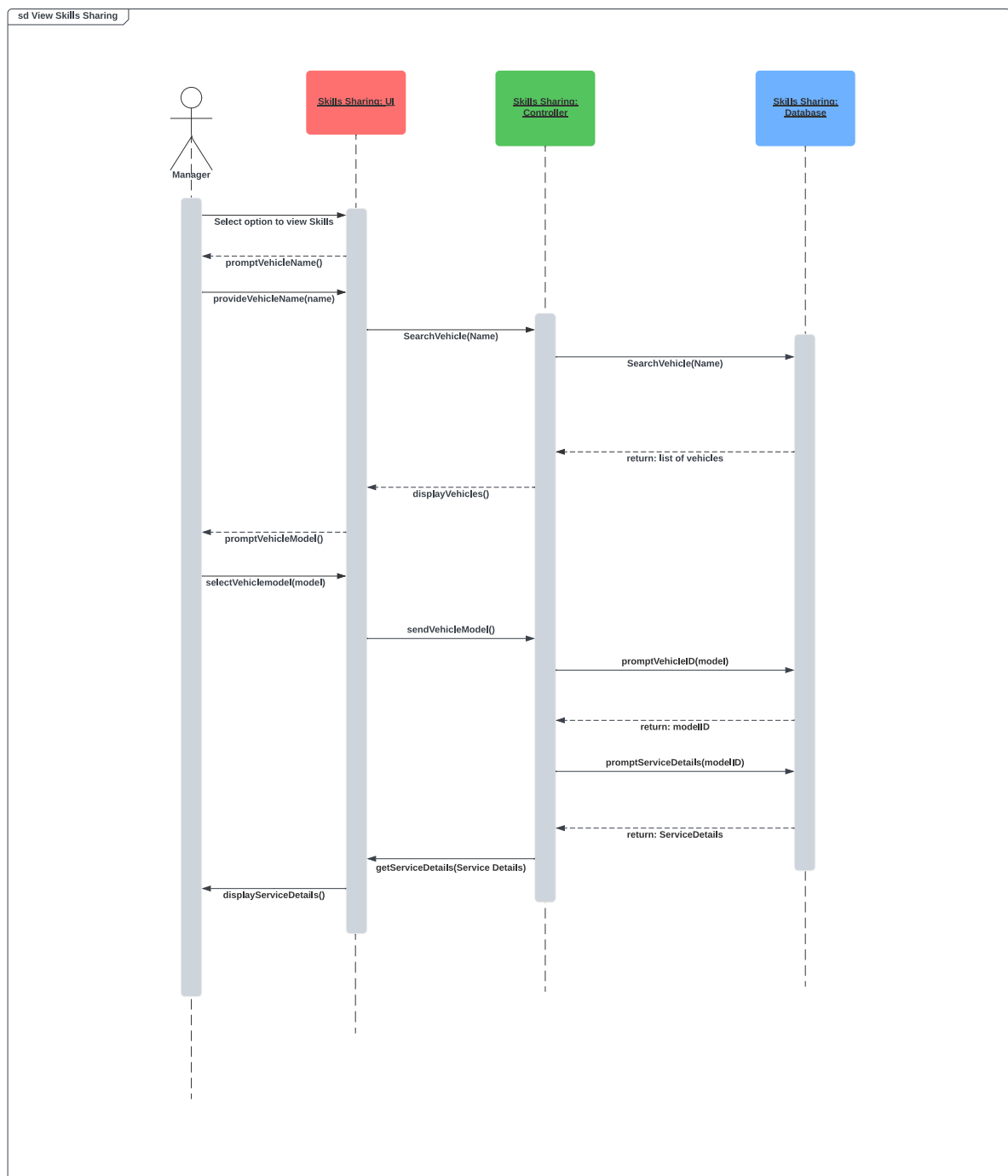


3.6 Skills Sharing

a) Add to Skill Sharing



b) View Skills Sharing



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